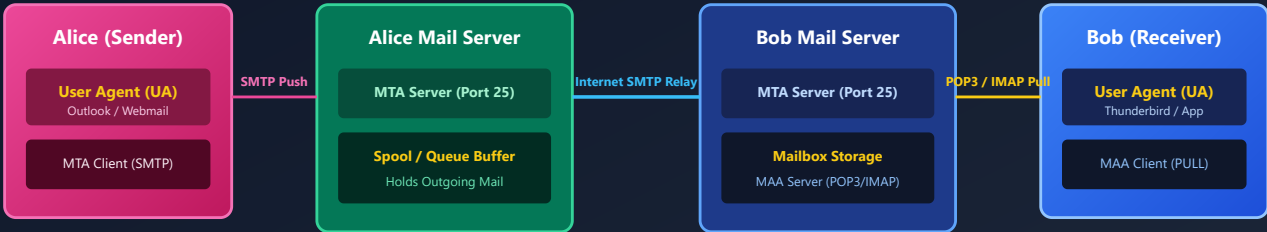


Module 04: Electronic Mail Architecture and MIME

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

ELECTRONIC MAIL (E-MAIL) 4-TIER ARCHITECTURE & MIME

User Agent (UA), Message Transfer Agent (MTA / SMTP), Message Access Agent (MAA / POP3 / IMAP) & MIME



1. Four E-Mail Scenarios

Scenario 1: Both on same timesharing host. • **Scenario 2:** Direct connection between 2 hosts. • **Scenario 3:** Alice remote, Bob local. • **Scenario 4 (Modern Web):** Both Alice & Bob connected to mail servers. Requires SMTP for PUSH + POP3/IMAP for PULL.

2. Core Architectural Roles • **User Agent (UA):** Client UI to compose, read, and organize email messages. • **Message Transfer Agent (MTA):** SMTP daemon pushing mail from sender to destination server. • **Message Access Agent (MAA):** POP3 / IMAP pulling mail from mailbox to recipient machine. 3. MIME (RFC 2045 - 2049) • **SMTP Limitation:** Can only send 7-bit ASCII. • **MIME Solution:** Encodes binary, audio, video into 7-bit ASCII before passing to SMTP. • **MIME-Version:** 1.0 • **Content-Type:** text, image, audio, multipart • **Transfer-Encoding:** Base64, Quoted-Printable

1. Electronic Mail 4-Scenario Evolutionary Framework

E-mail relies on a store-and-forward architecture transitioning from local single-system file append to a 4-tier distributed internet model.

The Fundamental Architectural Rule

SMTP is exclusively a **PUSH protocol** (sending client → server → server). Retrieving stored messages from a mailbox requires a **PULL protocol** (POP3 or IMAP4).

2. Functional Component Triad

Component	Abbr.	Role & Protocol Assignment
User Agent	UA	Client application interface for composing, reading, and managing mail (CLI: mailx, GUI: Outlook/Webmail).
Message Transfer Agent	MTA	Daemon responsible for relaying mail across the IP network using SMTP (Port 25) .
Message Access Agent	MAA	Client-server retrieval mechanism for pulling stored mail to client using POP3 (Port 110) or IMAP4 (Port 143) .

3. MIME Header Fields (RFC 2045)

Base64 Encoding Ratio: 3 raw bytes (24 bits) → 4 printable ASCII characters (33% size expansion)

- **MIME-Version:** Identifies protocol iteration (Standard value: 1.0).
- **Content-Type:** Declares media type and subtype (e.g., `multipart/mixed` , `image/jpeg`).

- **Content-Transfer-Encoding:** Declares transformation scheme (`base64` , `quoted-printable`).